

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date	13 July 2026
Team ID	
Project Name	Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Dashboard Visualization	Display interactive Tableau dashboard with key housing metrics Display KPI summary cards for total properties, average sale price, and total living/basement area
FR-2	Price & Feature Relationship Analysis	Analyse how sale price varies with bedrooms, bathrooms and living area Compare price distribution across overall grade / condition
FR-3	Renovation Impact Analysis	Display total sales distribution by years since renovation Show house-age distribution by renovation status
FR-4	House Age Analysis	Analyse house-age distribution by number of bedrooms, bathrooms and floors Compare age patterns across property segments
FR-5	Location-wise Price Analysis	Analyse sale price and property distribution by zipcode group Compare pricing patterns across locations
FR-6	Waterfront & View Premium Analysis	Analyse the price premium associated with waterfront-view properties Compare average price of waterfront vs. non-waterfront homes
FR-7	Interactive Filters	Apply filters by zipcode group, bedrooms, bathrooms and price range Enable drill-down into individual visualizations
FR-8	Tableau Story	Display a guided analytical story of housing-market trends Present key insights on renovation, age and bedrooms in a narrative format
FR-9	Public Access & Sharing	Publish the dashboard and story to Tableau Public for open access Enable sharing of the dashboard via a public link
FR-10	Business / Buyer Reporting	Present summarised housing-market insights for stakeholders Support data-driven pricing, buying and selling decisions

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The dashboard should be easy to use, with clear navigation and intuitive interactive visualizations for buyers, agents and analysts.
NFR-2	Security	The application should present housing analytics through Tableau Public without exposing personally identifiable buyer or seller data.
NFR-3	Reliability	The dashboard should consistently produce accurate and dependable analytical results from the underlying housing sales data.
NFR-4	Performance	Dashboard pages should load efficiently, and visualizations and filters should respond smoothly to user interaction.
NFR-5	Availability	The Tableau dashboard and story should remain accessible to users whenever housing-market insights are needed.
NFR-6	Scalability	The solution should be able to accommodate larger datasets and additional dashboards as the project grows in future.